ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. IT ENGINEERING — SEMESTER 4

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain HTML5 semantic elements and CSS3 Flexbox and Grid layouts with responsive design. [14 Marks]

[OR]

Q.1 (Alt) Explain JavaScript ES6 features: Promises, Async/Await, Arrow functions, and DOM manipulation. [14 Marks]

--- UNIT II ---

Q.2 Explain Node.js event loop, asynchronous I/O, and Express.js routing and middleware architecture. [14 Marks]

[OR]

Q.2 (Alt) Design RESTful APIs with CRUD operations and status codes (200, 201, 400, 401, 403, 404, 500). [14 Marks]

--- UNIT III ---

Q.3 Explain MongoDB NoSQL document modeling, indexing, and Mongoose aggregation pipelines. [14 Marks]

[OR]

Q.3 (Alt) Explain JWT (JSON Web Token) authentication and session security with cookies. [14 Marks]

--- UNIT IV ---

Q.4 Explain React.js component lifecycle, Virtual DOM, and Hooks (useState, useEffect, useContext). [14 Marks]

[OR]

Q.4 (Alt) Describe State Management in single page applications with Redux and Context API. [14 Marks]

--- UNIT V ---

Q.5 Explain Web Security practices: Cross-Origin Resource Sharing (CORS), CSP, and HTTPS. [14 Marks]

[OR]

Q.5 (Alt) Describe WebSockets for real-time bidirectional communication and progressive web apps (PWA). [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****